

TRACK: EDTECH • INTERNSHIP OPS

KNOWVIA

The Department-Scoped Knowledge & Operations Platform Built Specifically for
Multi-Department Tech Hub Programs.

Time Format: 3-Minute Presentation + 2-Minute Demo

Product Type: Progressive Web App (PWA)

Team: Sydonverse / Knowvia

The Chaos of Informal Channels

Why managing 50+ interns across WhatsApp and shared folders breaks down.

01 / DROWNED ANNOUNCEMENTS

Chat Noise & Missed Classes

Class links, reschedule notices, and curriculum updates get buried under hundreds of casual messages, memes, and banter.

02 / DELIVERABLE BLACKHOLE

Fragmented Submissions

Interns submit tasks across direct messages, random emails, and Google Drive links. Tutors lose track of who submitted what.

03 / SECURITY VULNERABILITY

Unvetted File Uploads

Open groups allow executable files, suspicious scripts, and nested archives without validation, risking security across the hub.

Field Discovery: What Surprised Us?

Direct findings from surveying Hub Administrators, Track Tutors, and Interns.

78%

of Interns reported missing crucial deadlines or meeting links due to chaotic group chat streams.

4 of 5

Tutors spent over 3 hours weekly manually chasing assignments across spreadsheets and DMs.

The Unexpected Revelation

"We initially assumed tech hubs needed a full enterprise LMS like Canvas or Blackboard.

What we learned: Hubs actively avoid enterprise LMSs because they are clunky, expensive, and alienating for fast-paced cohorts.

*What hubs actually needed was **the speed and familiarity of messaging** combined with **clean departmental boundaries, automated schedule alerts, and pre-storage file safety.***

Meet Knowvia

A high-performance Progressive Web App tailored for multi-department tech academies.

PILLAR 01

Department-Scoped Isolation

Frontend, Backend, UI/UX, and Cyber operate in strictly partitioned workspaces with zero cross-track distraction or noise.

PILLAR 02

Schedule & Milestone First

Dashboard-first next class countdown with agenda, real-time WebSocket rescheduling, and personal assignment progress meters.

PILLAR 03

Pre-Storage File Security

Multi-tier file scanner: blocks 23 executable extensions, detects double extensions, inspects ZIP headers, and stops malware pre-upload.

Live Product Walkthrough

2-Minute seamless live demonstration across Intern, Tutor, and Real-Time channels.

PHASE 1 • 0:00 – 0:40

Intern Dashboard

- Live upcoming class countdown & link.
- Dynamic milestone meter (Approved/Pending).
- Notification bell deep-link scrolling.

PHASE 2 • 0:40 – 1:20

Tutor Schedule & Uploads

- Real-time schedule editing modal.
- Instant WebSocket broadcast sync.
- Secure learning material file upload.

PHASE 3 • 1:20 – 2:00

Department Chat

- Scoped channel (zero foreign noise).
- Quoted reply threading.
- Deterministic initials avatar engine.

System Architecture & Tech Stack

Engineered for speed, strict security boundaries, and zero-cost cloud deployment.

CLIENT TIER (PWA)

React 19 + TypeScript + Vite: Ultra-fast responsive UI with a custom Vanilla CSS design system.
Service Worker (Cache v6) & Web Push: Installable on mobile/desktop; background automated class alerts.

APPLICATION LAYER

Node.js & Express API: Strict JWT role-scoping and department authorization guards.
Socket.io & File Heuristics: Room isolation by dept: {slug} with an in-memory ZIP central directory scanner.

DATA & CLOUD TIER

Supabase PostgreSQL with Row-Level Security (RLS): Prevents unauthorized cross-department data queries.
Prisma ORM & S3 Storage: Strict type-safety, connection pooling, and authenticated material streaming.

Market Analysis & The Knowvia Gap

Capturing the underserved intersection between chaotic chat and clunky enterprise LMSs.

INFORMAL APPS

WhatsApp / Telegram

- Free and familiar to everyone.
- Zero academic structure or grading.
- High noise; lost assignments.
- Files expire; malware risks.

ENTERPRISE LMS

Canvas / Blackboard

- Robust gradebooks & modules.
- Prohibitively expensive for hubs.
- Heavyweight, slow onboarding.
- Poor real-time messaging.

THE KNOWVIA ADVANTAGE

The Perfect Balance

- **Zero friction:** Installable PWA.
- **Department Isolation:** No track noise.
- **Integrated Core:** Schedule, assignments, file safety, and chat in one.
- **Addressable:** 150+ hubs & 3MTT centers.

Business Model & The Team

Sustainable B2B institutional monetization backed by proven product execution.

B2B SAAS REVENUE MODEL

Hub-Level Licensing

- **Hub Cohort License:** Annual subscription per cohort (up to 5 departments).
- **Enterprise Custom Tier:** Dedicated domain, branded UI, storage expansion & SLA.
- **High Margins:** 85%+ gross margin enabled by lightweight, modern cloud architecture.

TEAM & LIVE ACCESS

Ready for Deployment

Engineering Team: Frontend / PWA, Backend / RLS, UX & Research.
Verification: 88 / 88 Passing Automated Tests • Production Certified.

[SCAN QR TO TEST LIVE PWA ON MOBILE](#)

GitHub: Sydonverse / Knowvia